



Off the Books, Not Off the Shelf

Validating a Library Book-Swap Shelf's QR Circulation Tracker Against a Term of Twice-Daily Physical Counts

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Motivation

Facilities markets the book-swap shelf's QR tracker as a live circulation count, and a pending review cites its decline. Does the scanned count still track how often the shelf's books actually change hands?

Research questions

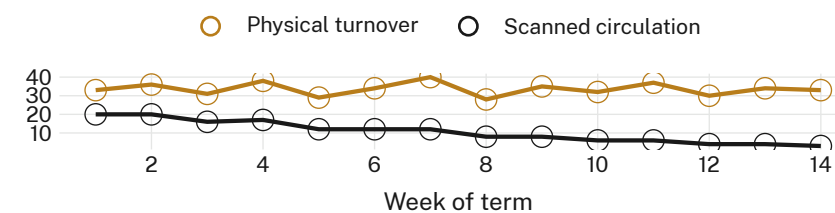
- Establish true turnover from a fixed schedule of physical counts.
- Compare scanned circulation against turnover, week by week.
- Test whether the undercount widens as the term progresses.

Study design

- 196 physical counts (2/day) · 14-week term · turnover from photos
- QR scan log pulled from the dashboard, aligned to weekly windows
- Sticker condition logged each count; restock weeks flagged for ablation

Findings

Scanned circulation fell from 61% of physical turnover in week 1 to 9% by week 14, while turnover itself held steady (median 33.5 books/week, IQR 31–36).



148 scans were logged against an estimated 470 books that changed hands across the term (31% overall capture); excluding the two post-restock weeks leaves the decline unchanged.

Discussion & conclusions

The undercount widens with its own hardware: most QR stickers were re-stuck, taped over, or missing by week 14, and the dashboard has no way to flag a stale code. A review reading the decline is reading the sticker, not the shelf. Next: a sticker-condition audit trigger, or counting the shelf directly.

Selected references

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“The sticker wears off. The shelf doesn’t.”

Scanned circulation fell from 61% to 9% of physical turnover across a 14-week term.